1 **(a)** A spring, which has an unstretched length of 0.650 m, is attached to a fixed point. A mass of 0.400 kg is attached to the spring and gently lowered until equilibrium is reached. The spring has then stretched elastically by a distance of 0.200 m.

Calculate, for the stretching of the spring,

(i) the loss in gravitational potential energy of the mass,

loss = ... J [1]

(ii) the elastic potential energy gained by the spring.

gain = ... J [2]

(b) Explain why the two answers to **(a)** are different.

.....

.....

..... [2]

(c) The mass on the spring is now set into simple harmonic motion by pulling it downwards by an additional 0.100 m and released from rest.

The angular frequency ω of a spring-mass system is given by $\omega = \sqrt{\frac{k}{m}}$, where k is the force constant and m is the mass of the load.

(i) Calculate the maximum speed of the mass,

maximum speed = ... m s^{-1} [2]

(ii) The velocity-displacement graph of the mass is shown in Fig. 1.1. Label on the graph the point **P** when the mass is first released and point **Q** when it first returns to its equilibrium position. Include numerical values on the axes. Take upwards to be positive. [1]

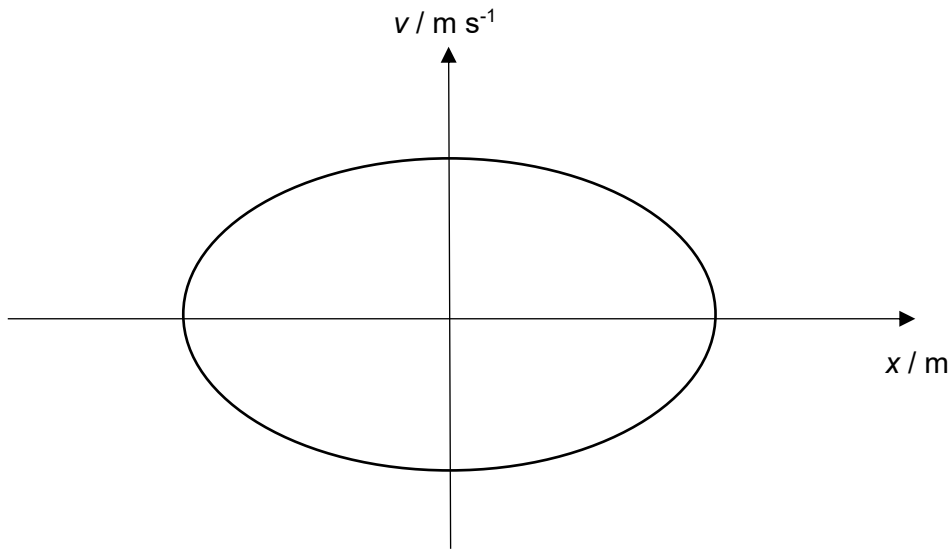

Fig. 1.1

(iii) The mass is now lowered into a container of water until it is completely submerged. The mass is again displaced downwards from its equilibrium position by 0.100 m and released from rest. Throughout the oscillations, the mass remains under water.

Sketch on Fig. 1.1 the variation of velocity with respect to displacement for one complete oscillation of the mass. [2]